llmXive Automated Review of Zone of Proximal Policy Optimization: Teacher in Prompts, Not Gradients

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and their support by the provided evidence.

1. Logical Inconsistency in Teacher Capability Claims (Section 5.3) The manuscript states in Section 5.3: "BCQ candidates dry up as student scales (teacher also fails)." This phrasing is factually misleading. The teacher model (Qwen3.5-27B) is fixed; its performance on a static benchmark does not change based on the student's scale. The intended meaning is likely that as the student improves (scales up), the set of problems where the *teacher succeeds* and the *student fails* (the "Zone of Proximal Development") shrinks because the student begins to solve problems the teacher can also solve. The current phrasing incorrectly attributes the reduction in candidates to the teacher failing more often as the student scales. This should be rephrased to clarify that the *gap* between teacher and student narrows, reducing the available BCQ opportunities, not that the teacher's performance degrades.

2. Precision of Distillation Impact Claims (Introduction) The Introduction claims: "Distillation degrades students by -2.5/-1.8/-0.9/-0.3 pp" on LLM and Video benchmarks. This is a strong negative claim. A review of the provided tables (e.g., Tab. 1, Tab. 2, and the ablation tables) is necessary to verify if *all* distillation methods (Off-policy and On-policy) consistently show these specific negative deltas compared to the Base model. If the "Base" model is the reference, and the distillation methods show mixed results (some positive, some negative), attributing a uniform negative degradation to "Distillation" as a monolith is an overgeneralization. The text should specify which distillation variant (e.g., Off-policy vs. On-policy) is responsible for these specific negative numbers, or if the average of both is being reported. If the numbers are averages, the variance should be acknowledged to avoid implying a universal failure of distillation.

3. Semantic Precision in Performance Comparisons (Section 5.2) In Section 5.2, the text claims the 9B ZPPO student "approaches" the 27B teacher on AIME25 (70.0 vs 70.0) and is "within <= 1.0 pp" on OCR_EN (56.7 vs 55.7). While the numbers in the tables (Tab. 12) support these values, the narrative "approaches" for AIME25 is accurate (exact match), but for OCR_EN, the student actually *surpasses* the teacher by 1.0 pp. Using "approaches" for a case where the student exceeds the teacher might be semantically imprecise, though not factually false. However, the claim that the teacher is "least saturated" on HLE (16.0) and VBlind is supported by the low scores, but the text should ensure it doesn't imply the teacher *cannot* solve these, only that the current benchmark is hard for the teacher as well.

4. Citation Consistency The paper cites `vygotsky1978mind` for the "zone of proximal development" concept, which is appropriate. The citations for GRPO (`shao2024deepseekmath`), DAPO (`yu2025dapo`), and REINFORCE++ (`hu2025reinforce++`) are consistent with the method descriptions. No obvious citation mismatches were found where a source is cited for a claim it does not support, provided the external papers (which are not fully visible here) contain the described methods. The internal consistency of the method description (e.g., DAPO's asymmetric clip values) matches the cited values in the text.

Conclusion The paper's core claims are generally supported by the data, but the phrasing regarding the teacher's failure rate (Section 5.3) is logically flawed and requires correction. Additionally, the generalization of distillation's negative impact needs more precise attribution to specific methods to avoid overstatement.

Action items.

- [**writing**] Section 5.3 claims 'BCQ candidates dry up as student scales (teacher also fails).' This is factually incorrect; the fixed 27B teacher's failure rate is constant. Clarify that the gap narrows as the student improves, reducing the pool of hard questions where the teacher succeeds but the student fails.
- [**science**] Introduction claims 'Distillation degrades students by -2.5/-1.8...' on LLM/Video. Verify if this specific negative delta applies to all distillation variants or just one. If it's an average, specify this to avoid overgeneralizing that all distillation methods universally degrade performance compared to the base.
- [**writing**] Section 5.2 states the 9B student 'approaches' the teacher on OCR_EN (56.7 vs 55.7). Since the student actually surpasses the teacher by 1.0 pp, use 'surpasses' or 'exceeds' for accuracy. Ensure 'approaches' is reserved for cases where the student is strictly below or equal to the teacher.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure fails to visualize the specific error described in the caption (snapping 500 to 400) and lacks the comparative traces (BCQ vs NCQ) implied by the text, presenting only a single generic line graph.

Figure 2

The rendered image for Figure 2 is a photograph of groceries that is entirely unrelated to the caption's description of a visual counting task involving model traces and rollouts. The figure fails to support the paper's claims as it contains no data, text, or visualizations relevant to the described experiment.

Figure 3

The provided image is a photograph of a room interior and does not match the figure caption, which describes a technical example of a 4B student model's performance on a visual counting task. The figure fails to display the described traces, data, or reasoning process.

Figure 4

The provided image for Figure 4 is a photograph of a person with donuts, which does not match the caption's description of a scene QA example involving a coat. This is a critical mismatch between the visual content and the scientific claim.

Figure 5

The figure is a mismatch for its caption; it shows a slice of pizza instead of the described image containing straws required to evaluate the visual counting task.

Figure 6

Figure 6 effectively illustrates the conceptual failure modes and ZPPO mechanisms using clear diagrams, but the caption is incomplete as it fails to explicitly describe the third failure mode ('New Problems') shown in panel (a) and does not map the technical descriptions to the specific sub-panels.

Figure 7

The figure provides a clear visual overview of the ZPPO framework, but the captions for panels (a), (b), and (c) describe the processes in a way that does not fully align with the rendered content, which focuses on prompt templates and high-level flow rather than the detailed processes described.

Figure 8

The figure is readable and clearly compares the two methods, but the caption's description of the y-axis as a ratio contradicts the axis label and raw count values shown in the plot. Additionally, the symbol used for the GRPO method in the legend (†) does not match the symbol in the caption ($\hat{}$).

Figure 9

The figure presents the requested data but contains a rendering error in the legend text for 'Easy' and a mathematical inconsistency in the stacked bar chart where the percentages for the 0.8B model sum to more than 100%.

Figure 10

Figure 10 is scientifically clear with well-labeled axes and data points, but the layout is slightly cluttered due to the unboxed legend floating above the plots and the repetition of x-axis labels across the top row.

Figure 11

The figure effectively visualizes performance gains across teacher sizes with clear data labels and a readable legend. However, the caption contains a broken LaTeX placeholder (‘$$’) that obscures the specific method being evaluated, and the y-axis relies on the caption to define the delta symbol.

Figure 12

The figure effectively visualizes the evolution of replay buffer composition across model scales, but the legend title and y-axis label lack the specific context (‘at admission’, ‘total occupancy’) provided in the caption, which could lead to misinterpretation of the data categories.

Action items.

- [**science**] Figure 1: The caption claims the student ‘down-snaps’ the unlabelled y=500 at x=1 to the 400-row, but the plot shows a line passing exactly through the grid intersection (1, 400) with no visual indication of the ‘unlabelled 500’ point or the snapping error described.
- [**science**] Figure 1: The caption describes a ‘BCQ/NCQ example’ contrasting two traces, yet the figure displays only a single linear plot with no visual distinction between the ‘400-row chain’ and the ‘exact-intersection’ chain mentioned.
- [**writing**] Figure 1: The plot includes generic axis labels ‘x’ and ‘y’ at the arrowheads which are redundant and clutter the visualization given the specific ‘Time (hours)’ and ‘Distance (miles)’ labels already present.
- [**fatal**] Figure 2: The rendered image displays a photograph of groceries (bananas, apples, bread, salt, stuffing mix) on a table, which does not match the caption’s description of a ‘Visual counting’ example involving ‘BCQ/NCQ’ traces, ‘teacher-correct’ and ‘student-wrong’ rollouts, or any textual data.
- [**science**] Figure 2: The image content (a still life of food items) is completely unrelated to the paper’s subject matter (ZPPO, LLM training, policy optimization) and fails to provide any evidence for the claims made in the caption regarding model performance or reasoning traces.
- [**fatal**] Figure 3: The rendered image is an interior design photograph of a console table with vases, which is completely unrelated to the caption’s description of a ‘Visual counting, 4B student’ example involving BCQ/NCQ traces and text-based reasoning.
- [**fatal**] Figure 4: The rendered image is a photograph of a person holding a box of Krispy Kreme donuts, which is completely unrelated to the caption’s description of a ‘Scene QA’ task involving a ‘2B student’ analyzing a ‘coat’ with ‘braided-cord’ and ‘pocket flap’ features.
- [**fatal**] Figure 5: The rendered image displays a slice of pizza on a paper plate, which contradicts the caption’s description of a ‘Visual counting’ task involving a ‘cluster of straws’ and a ‘right-edge cluster’.
- [**science**] Figure 5: The image content (pizza) does not support the claims regarding the

student's failure to count straws or the BCQ/NCQ prompt mechanics described in the caption.

- **[writing]** Figure 6: The caption describes panel (a) as 'Two failure modes' but the panel contains three distinct sub-panels (1, 2, 3) labeled 'Distillation', 'RL from Teacher', and 'New Problems'; the caption should explicitly list all three modes to match the visual content.
- **[writing]** Figure 6: Panel (a) sub-panels 1 and 2 use the terms 'Distillation' and 'RL from Teacher' which are not defined in the caption, while the caption text refers to 'fitting the student to a much larger teacher's logits' and 'injecting a teacher response into the policy gradient' without explicitly linking these descriptions to the specific sub-panels.
- **[writing]** Figure 7: The caption describes panel (a) as 'Hard questions... admitted to the prompt replay buffer', but the rendered panel (a) is titled 'Rollouts and Hard Question Admission to Replay Buffer' and depicts the admission logic (Teacher vs Student rollouts) rather than the buffer composition itself, creating a mismatch between the caption's description and the visual content.
- **[writing]** Figure 7: The caption describes panel (b) as 'BCQ pairs one correct teacher response with one wrong student response', but the rendered panel (b) is titled 'Binary Candidate-included Question (BCQ)' and shows a prompt template with one correct and one wrong candidate, which is a representation of the prompt format rather than the pairing process described.
- **[writing]** Figure 7: The caption describes panel (c) as 'NCQ aggregates the student's wrong rollouts into a single prompt', but the rendered panel (c) is titled 'Negative Candidate-included Question (NCQ)' and shows a prompt template with multiple wrong candidates, which is a representation of the prompt format rather than the aggregation process described.
- **[writing]** Figure 7: The caption describes panel (d) as 'Integrated batch drives the policy gradient update with RL Recipe', but the rendered panel (d) is titled 'Overview of ZPPO' and shows the training loop with 'BCQ' and 'NCQ' labels in the batch, which is a high-level overview rather than a detailed description of the policy gradient update.
- **[writing]** Figure 8: The caption defines the y-axis as 'Cumulative graduate counts (graduated/admitted = ratio)', but the axis label in the plot is simply 'cumulative graduations' and the tick marks (0, 200, ..., 1000) represent raw counts, not the ratios described in the caption.
- **[writing]** Figure 8: The legend uses the symbol '†' for 'Qwen3.5-2B + GRPO†', but the caption uses the symbol '^' for 'GRPO^'; the symbols do not match.
- **[writing]** Figure 9: The caption for (a) defines 'Easy' as $r_x > 0.5$, but the rendered legend uses a double exclamation mark (r_x!!0.5) which is a rendering error or typo.
- **[science]** Figure 9: The stacked bar chart in (a) shows percentages (e.g., 36%, 38%, 34%) that sum to 108% for the 0.8B model, indicating a calculation or labeling error in the data visualization.

- [**writing**] Figure 9: The x-axis label 'Fraction of Rollout over the entire training' is grammatically awkward and imprecise; 'Fraction of batch composition' would be clearer.
- [**writing**] Figure 10: The legend at the top of the figure is not enclosed in a box or panel, making it visually float above the subplots and potentially confusing regarding which data series it applies to.
- [**writing**] Figure 10: The x-axis label 'Iterations per step l' is repeated for every subplot in the top row (a, b, c) rather than being shared or placed only on the bottom row, creating visual clutter.
- [**writing**] Figure 11: The caption 'All-averaged gain

$$(pp)'containsabrokenLaTeXplaceholder($$

) instead of the method name (e.g., ZPPO), making the metric definition unclear.
- [**writing**] Figure 11: The y-axis label 'Avg Δ (pp)' uses the symbol Δ without explicitly defining it as 'gain' or 'improvement' in the axis label itself, relying on the caption which is currently malformed.
- [**writing**] Figure 12: The legend title 'Entry Rollout Accuracy' is ambiguous; the caption clarifies these are 'admission bins' (entry accuracy at admission), but the legend lacks the word 'admission' or 'at admission' to distinguish them from current training accuracy.
- [**writing**] Figure 12: The y-axis label '# entries in buffer' is generic; the caption specifies 'buffer occupancy', but the axis label does not explicitly state that the stacked height represents the total buffer size limit (10,000).

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific acronyms and jargon that are not defined at their first point of use, creating unnecessary friction for non-specialist readers. The most critical omissions are the acronyms **BCQ** (Binary Candidate-included Question) and **NCQ** (Negative Candidate-included Question), which are central to the proposed method. These terms appear in the Introduction and Section 3 without their full expansions, forcing the reader to search the text to understand what the "B" and "N" stand for. Similarly, **VLM** (vision-language model) is used repeatedly in the Introduction without definition.

In the experimental setup and baseline descriptions, the acronym **JSD** (Jensen–Shannon divergence) is used without expansion. Additionally, **FLOPs** appears in the compute cost analysis without being spelled out. While terms like "on-policy" and "zero-advantage" are standard in reinforcement learning literature, their usage in the Introduction assumes a level of prior knowledge that excludes readers from adjacent fields. The paper would benefit significantly from a "glossary-style" approach in the first paragraph of the Introduction or the first mention of each term, ensuring that every acronym is explicitly defined before being used as a shorthand. This is a straightforward fix that would greatly improve the paper's accessibility without altering its scientific content.

Action items.

- [**writing**] The manuscript relies heavily on domain-specific acronyms and jargon that are not defined at their first point of use, creating unnecessary friction for non-specialist readers. The most critical omissions are the acronyms BCQ (Binary Candidate-included Question) and NCQ (Negative Candidate-included Question), which are central to the proposed method. These terms appear in the Introduction and Section 3 without their full expansions, forcing the reader to search the text to understand what the "B

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent logical structure where the problem (zero-advantage groups in RL and brittleness in distillation) motivates the solution (ZPPO). The core premise—that keeping the teacher in the prompt preserves on-policy gradients while leveraging teacher knowledge—is logically sound and consistently applied throughout the method description.

However, there are gaps in the logical derivation of specific causal claims:

1. **Super-additivity Claim:** The assertion that the combination of BCQ, NCQ, and the Replay Buffer is "super-additive" (Sec. 4.3) is supported by the empirical data in Table 5 (ZPPO > sum of parts), but the *mechanism* for this super-additivity is not logically derived. The paper states the combination "compounds far beyond isolated effects" but does not explain *why* the interaction between the buffer and the reformulated prompts creates a non-linear gain. Is it due to the specific curriculum of hard questions? The logical link between the components and the non-linear gain is asserted but not rigorously demonstrated.
2. **NCQ Mechanism Validity:** The logical consistency of the NCQ mechanism is challenged by the 0.8B results. The premise is that listing wrong answers helps the student avoid them. The data (Table 12) shows a 82.7% failure rate (match-neg) for the 0.8B model, meaning the mechanism effectively fails for the smallest scale. The conclusion that NCQ is a generalizable component for the "Zone of Proximal Development" is logically weakened by this scale-dependent failure, which the paper acknowledges but does not fully reconcile with the general claim of NCQ's utility.
3. **Distillation Causality:** The claim that distillation "hurts generalization" due to "mem-

orization" is a strong causal statement. While the results show a performance drop on LLM/Video benchmarks, the paper does not provide a logical derivation or intermediate evidence (e.g., analysis of feature representations or loss landscapes) that directly links the distillation objective to the specific degradation on out-of-distribution tasks. The conclusion follows the data, but the *causal mechanism* remains an assumption rather than a proven logical step.

The paper is logically consistent in its high-level narrative but requires stronger mechanistic justification for its specific causal claims regarding super-additivity and the universal applicability of the NCQ mechanism.

Action items.

- [**science**] The 'super-additive' claim (Sec 4.3) lacks a derived mechanism. Data shows ZPPO > components, but the non-linear interaction logic is asserted, not proven.
- [**science**] NCQ's 'avoidance' mechanism fails at 0.8B (82.7% match-neg rate, Tab 12). The conclusion that NCQ is universally robust contradicts this scale-dependent failure data.
- [**science**] The causal link between 'logit matching' and 'generalization failure' is asserted but not derived. No mechanistic evidence links distillation to the specific OOD degradation observed.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the universality and magnitude of ZPPO's improvements that slightly exceed the statistical and empirical evidence provided.

First, the claim in Section 5.2.2 that a 9B ZPPO student "approaches the 27B teacher" is an over-generalization based on a single benchmark (AIME25). While the scores match (70.0 vs 70.0), the student lags significantly on other critical benchmarks like HLE (9.8 vs 16.0) and VBlind. The text should be tempered to state that the student matches the teacher on *specific* benchmarks where the teacher is not saturated, rather than implying a general convergence of capability.

Second, the "Finding" box in the Introduction and Section 5.1 asserts that distillation "hurts generalization" while ZPPO improves it. While the macro-averages support this, the cluster bootstrap confidence intervals in Table 10 (Appendix) reveal that for Video benchmarks at 4B and 9B, the difference between ZPPO and the best non-ZPPO baseline is not statistically significant (CIs include zero: [-0.24, +0.90] and [-0.02, +0.86]). Claiming a definitive "hurt" vs "improve" dichotomy without qualifying these specific non-significant regimes overstates the robustness of the finding.

Finally, the description of the BCQ/NCQ combination as "super-additive" (Section 3.4, Section 5.3) suggests a consistent, compounding effect across all scales. The ablation results (Tables 7 and 8) show that for the smallest model (0.8B) on Video benchmarks, the incremental gain of adding both components is minimal compared to the buffer-only baseline. The term "super-additive" should be qualified to reflect that this phenomenon is primarily observed in the VLM and LLM domains and at larger student scales, rather than being a universal property of the method.

Action items.

- [**writing**] The claim that ZPPO 'approaches' the 27B teacher on AIME25 (70.0 vs 70.0) in Sec. 5.2.2 is an over-interpretation of a single data point. The 9B student matches the teacher on one specific benchmark but lags significantly on others (e.g., HLE: 9.8 vs 16.0). The text should qualify this as 'matching on specific benchmarks' rather than a general approach to teacher capability.
- [**science**] The assertion that 'Distillation hurts generalization' (Finding 1) is slightly overstated given the confidence intervals in Tab. 10 (Appendix). For Video benchmarks at 4B/9B, the CI for ZPPO vs Best-non-ZPPO includes zero ([-0.24, +0.90] and [-0.02, +0.86]). The conclusion should acknowledge that the superiority of ZPPO over all baselines is not statistically significant in these specific high-scale video regimes.
- [**writing**] The paper claims BCQ/NCQ are 'super-additive' (Sec. 3.4, Sec. 5.3) based on macro-averaged gains. However, the ablation tables (Tab. 7, Tab. 8) show that for the 0.8B model on Video benchmarks, the gain from BCQ+NCQ is marginal compared to GRPO+Buffer. The term 'super-additive' implies a consistent multiplicative effect across all scales and domains, which the data does not uniformly support. The claim should be restricted to the specific VLM/LLM regimes where the effect is strong.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript addresses safety and ethics primarily in Section 6 ("Ethical Considerations") and the "Limitations" section. The authors correctly identify that ZPPO inherits upstream biases from the base Qwen3.5 models and explicitly state that the method targets correctness rather than safety, requiring upstream safety alignment. This is a responsible and accurate assessment of the method's scope.

Regarding dual-use risks, the paper focuses on improving reasoning capabilities in Vision-Language Models (VLMs) and LLMs on benchmarks like MMLU, AIME, and various VQA tasks. While enhanced reasoning capabilities can theoretically be misused, the paper does not introduce novel capabilities for generating harmful content, bypassing safety filters, or automating cyber-attacks. The training data (ZPPO-77K) is derived from public datasets

(Vero-600k, MMFineReason-SFT-586K) and the evaluation benchmarks are standard academic resources. There is no indication of sensitive or private data usage; the authors cite HuggingFace datasets which are generally public.

The methodology involves Reinforcement Learning from Human Feedback (RLHF) variants (specifically GRPO) and knowledge distillation techniques. The "teacher" model is a frozen 27B parameter model, and the "student" models are smaller variants. The process does not involve human-in-the-loop data collection that would raise IRB/IACUC concerns, as the "judges" are either rule-based graders or the teacher model itself (LLM-as-a-judge). The use of LLMs as judges is standard in the field and does not constitute human subject research in this context.

The paper includes a "Limitations" section that acknowledges the "Teacher-bounded zone" (i.e., if the teacher fails, the method cannot help) and the potential tension with dynamic sampling. This transparency regarding the method's boundaries is a positive ethical practice. The authors do not make overblown claims about the safety of the resulting models, instead deferring to upstream alignment.

No conflicts of interest are apparent in the provided text, though the authors are affiliated with major research institutions (NVIDIA, University of Toronto, etc.), which is standard. The paper does not appear to facilitate the creation of dangerous agents or the exploitation of vulnerabilities. The focus on mathematical and visual reasoning benchmarks limits the immediate risk of generating harmful text compared to methods focused on open-ended generation or social engineering.

In summary, the paper demonstrates appropriate awareness of its ethical boundaries, does not utilize sensitive data, and does not present significant dual-use risks beyond the general capabilities of the underlying foundation models. The safety considerations section is adequate for the scope of the work.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a novel method (ZPPO) for improving small VLMs by keeping the teacher in the prompt rather than the gradient, addressing the zero-advantage problem in standard RL. The experimental design is generally robust, utilizing a large suite of 31 benchmarks across VLM, LLM, and Video domains, and comparing against strong baselines including distillation and standard GRPO variants. The inclusion of a cluster bootstrap to assess benchmark-selection robustness is a strong methodological choice.

However, the scientific evidence regarding the statistical significance of the reported gains requires clarification. The cluster bootstrap (Sec. 5.4) effectively addresses the variance introduced by the specific choice of benchmarks but does not account for the inherent stochasticity of the RL training process itself (e.g., random rollouts, initialization noise). The paper reports results from single training runs (implied by the lack of "mean ± std" notation in the main tables). For RL methods, where performance can vary significantly between seeds, a single run is often insufficient to claim a definitive improvement, especially for smaller effect sizes (e.g., the +0.3 to +0.4 pp gains on Video benchmarks). The confidence intervals provided are conditional on the benchmark set, not the training run.

Furthermore, the mechanism of BCQ relies on the student correctly discriminating between a teacher trace and a student trace. The audit in Table 8 shows BCQ accuracy is only 36-69%, meaning the student frequently fails to identify the correct candidate. While the authors argue this is evidence against trivial matching, the paper does not explicitly analyze the impact of these failures on the gradient update. If the student consistently picks the wrong candidate, does the method introduce a negative bias? The current evidence supports the *existence* of a signal but lacks a rigorous analysis of the *noise* introduced by the imperfect discrimination, which is critical for validating the "Zone of Proximal Development" claim.

Finally, the claim of "super-additive" effects (ZPPO > GRPO + BCQ + NCQ + Buffer) is supported by point estimates in the ablation tables (e.g., Tab. 9), but the interaction terms are not statistically tested. Given the variance in RL, the observed super-additivity could potentially be an artifact of a single lucky run. Replication with multiple seeds or a more rigorous statistical test for interaction effects would strengthen the central claim.

Action items.

- [**science**] The cluster bootstrap analysis (Sec. 5.4, Tab. 10-11) resamples benchmarks to assess robustness to benchmark selection but does not account for run-to-run training variance. Given the stochastic nature of RL training (rollouts, advantage estimation), the reported 95% CIs may underestimate total uncertainty. Please clarify if single-run results are sufficient or if multiple seeds were averaged.

- [**science**] The BCQ audit (Tab. 8) reports accuracy of 36-69%, yet the method relies on the student correctly identifying the teacher's trace. The paper does not explicitly quantify the false-positive rate (student selecting the wrong candidate) or analyze if the 'correct' teacher trace is actually optimal for the student's current capability, which could introduce noise into the gradient.

- [**science**] The claim of 'super-additive' gains from combining BCQ/NCQ with the replay buffer (Sec. 5.3) is supported by ablation tables, but the interaction effect size is not statistically tested (e.g., via ANOVA or interaction terms). The current evidence relies on point-estimate differences which may not be robust to the variance inherent in RL training.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in the paper is generally rigorous in its use of benchmark-level cluster bootstrapping to establish the robustness of macro-average gains (Sec. 5.6, Tabs. 12-13). The authors correctly identify that benchmark selection is a primary source of variance in LLM evaluation and address it by resampling benchmarks 10,000 times. The resulting 95% CIs effectively demonstrate that the ZPPO improvements over baselines (GRPO†, Off-Distill) are not artifacts of a specific benchmark subset, as the intervals exclude zero in nearly all cases.

However, there are three specific areas where the statistical treatment requires clarification or strengthening:

1. **Run-to-Run Variance vs. Benchmark Variance:** The cluster bootstrap (Sec. 5.6) explicitly quantifies uncertainty due to *which* benchmarks are included in the average. However, the paper appears to report results from a single training run per configuration (implied by the lack of "mean ± std" notation in the main tables). In RL, training dynamics can be highly sensitive to initialization and sampling noise. By only bootstrapping the benchmarks, the reported CIs may underestimate the total uncertainty if the single run is an outlier. The authors should either report multiple seeds (e.g., n=3) to estimate run-to-run variance or explicitly state that the single-run results are representative and that the benchmark bootstrap is the only source of reported uncertainty.
2. **Statistical Significance of Scale-Dependent Effects:** In the BCQ/NCQ audit (Sec. 5.7, Tabs. 14-15), the authors observe a dramatic drop in "match-neg" (repeating a wrong answer) for the NCQ task as model scale increases (82.7% at 0.8B vs. 0.2% at 9B). While the trend is clear, the paper treats these as descriptive statistics. To support the claim that this is a genuine "collapse" of the failure mode rather than random variation, a formal statistical test (e.g., Fisher's exact test or a chi-square test for trend) comparing the match-neg rates across the four scales should be included.
3. **Uncertainty in Ablation Interactions:** The central claim of "super-additivity" (ZPPO > BCQ + NCQ + Buffer) relies on the difference between the full method and the sum of its parts (Tab. 11). The table presents only point estimates. Given the magnitude of the gains, it is crucial to know if the interaction term is statistically significant. Providing confidence intervals for the ablation differences or conducting a test for the interaction effect would strengthen the evidence for the super-additive claim.

Overall, the use of benchmark bootstrapping is a strong methodological choice, but the lack of run-level variance estimation and formal significance testing for the component audits leaves some statistical gaps.

Action items.

- **[science]** The cluster bootstrap (Sec. 5.6) resamples benchmarks to assess macro-average robustness but does not account for run-to-run variance. Given the small number of training runs (implied n=1 per method/scale), the authors should clarify if the reported CIs underestimate total uncertainty or provide a justification for treating the single run as representative.
- **[science]** In the BCQ/NCQ audit (Tab. 14-15), the 'match-neg' metric for NCQ at 0.8B is

82.7%. The text attributes this to 'limited capacity,' but the statistical significance of this scale-dependent drop (from 82.7% to 0.2%) is not tested. A formal test (e.g., chi-square or Fisher's exact) comparing match-neg rates across scales is needed to support the claim of a 'collapse'.

- [**science**] The ablation study (Tab. 11) reports point estimates for component contributions (BCQ vs. NCQ) without confidence intervals. Given the super-additive claim, the authors should provide error bars or statistical tests to confirm the interaction effect is not due to random fluctuation in benchmark selection.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high standard of technical writing, with a clear logical flow from the problem statement (zero-advantage groups in RL) to the proposed solution (ZPPO). The prose is generally concise, and the distinction between the proposed method and baselines is articulated effectively. The use of active voice in the methodology section aids readability.

However, there are minor areas where precision could be improved to prevent ambiguity. In Section 5, the description of BCQ candidate availability ("dry up") is slightly colloquial and could be misinterpreted as a data scarcity issue rather than a teacher capability limitation. Additionally, the introduction could benefit from a more immediate operational definition of the "zone" to ground the Vygotsky analogy for a broader ML audience. Finally, the limitations section contains a sentence fragment regarding the "hybrid approach" that disrupts the grammatical flow and obscures the proposed solution's mechanism. Addressing these specific phrasing issues will enhance the overall clarity and professional polish of the paper.

Action items.

- [**writing**] In Section 5 (Experiments), the phrase 'BCQ candidates dry up as student scales' is ambiguous. It is unclear if this refers to the *availability* of correct teacher candidates or the *frequency* of their usage. Clarify whether the teacher model fails on harder benchmarks as scale increases, reducing the pool of valid BCQ pairs.

- [**writing**] The abstract and introduction use the term 'zone of proximal development' without immediately defining the specific 'zone' boundaries for the algorithm (e.g., mean rollout accuracy < 0.5). While defined later, a brief parenthetical definition in the first mention would improve flow for non-psychology readers.

- [**writing**] In the 'Limitations' section, the sentence 'ZPPO retains all-wrong questions; dynamic sampling deletes them' lacks a clear subject-verb connection to the proposed 'hybrid approach.' Rephrase to explicitly state how the hybrid approach resolves this tension (e.g., 'A hybrid approach is proposed to retain hard questions while dynamically pruning those that remain unsolvable').